XRPL Validator — Daily Engineering Report · 2026-08-17

Period: 2026-08-17 00:00 to 24:00 US Eastern (2026-08-17 04:00 to 2026-08-18 04:00 UTC). Generated 2026-10-21 00:47 EDT by report generator v3.0. Sources: the XRP Ledger (ledger headers and amendments, read from a reference rippled node), the validator's signing log and receipt logs, the deployment pipeline's gate, soak and health records, the git history of xrpl-validator and the XRPL Rust SDK, and data.xrpl.org. Scope: XRPL work only.

Revised on 2026-10-21 in report format v3; the original is kept in this repository's history. Built only from records that exist for this day. Available: xrpl-validator merge records (they begin 2026-07-04). Not available: ledger headers (the reference node holds ledgers from 2026-10-04 22:52 UTC); signing records (they begin 2026-09-19); Step 1 receipt records (they begin 2026-08-29); Steps 2–5 receipt records (they begin 2026-10-09); deployment records (they begin 2026-09-11); amendment data; data.xrpl.org agreement scores (scored days begin 2026-04-04; some days have none).

Status

Not assessed — no signing records for this day.

Criteria. Normal: coverage outside planned maintenance at or above the 99.9% objective, every validation agreed with the network's validated ledger, and no engine receipt is open. Degraded: coverage below the objective, or an open engine receipt. Impaired: any validation that disagreed with the network, or coverage below 90%.

Executive summary

- **Validator:** no signing records for this day (they begin 2026-09-19).
- **Native engine:** no receipt records for this day (they begin 2026-08-29).
- **Changes:** 0 deployments, 0 commits merged into main.

Service levels

No signing records for this day (they begin 2026-09-19).

Native engine conformance

Earlier FFI shadow log (libxrpl applied against the validator's mirror; superseded by Step 1): 7,437 transaction-level differences recorded this day, across 142 ledgers. These were not triaged into receipts.

No receipt records for this day (Step 1 records begin 2026-08-29, Steps 2–5 records 2026-10-09).

Live counters are kept only for the current run; none exist for a past day.

Receipt register

No receipt records for this day (Step 1 records begin 2026-08-29).

Incidents

Not assessed: no signing records for this day.

Change log

Deployments

No deployment records before 2026-09-11.

Merged into main

No merges into main.

Findings fixed

No engine findings merged into main this day.

Other engineering changes

- **fix(xrpl-ledger): a terminal issuer-only change ripples between gateways, never a book** (`04d6c8b`, xrpl-validator) toStrand appends a terminal order book for the delivered asset only when the CURRENCY changes (PaySteps.cpp:220-233); its own comment is explicit — "for offer crossing (only) we do use an offer book even if all that is changing is the Issue.account". A PAYMENT whose last hop already holds the delivered currency under a DIFFERENT issuer therefore gets no book at all: the delivery issuer enters normPath as an ACCOUNT …
- **fix(xrpl-ledger): the balance clamps the GROSS, and a pass that misses limitQuality is discarded whole** (`23f1153`, xrpl-validator) Two coupled rules behind an IoC cluster one bot repeated every ledger — 11 ledgers, 24 divergences, 77% of everything open. Neither rule fixes it alone.
- **fix(xrpl-ledger): a pseudo-account cannot receive an escrow** (`8e48764`, xrpl-validator) `EscrowCreate::preclaim` refuses a pseudo-account destination with tecNO_PERMISSION immediately after the tecNO_DST read, BEFORE the token helper and before the tag test (EscrowCreate.cpp:350-352). It is deliberately not amendment-gated, because every write to a discriminator field is.
- **fix(xrpl-ledger): the strand upper bound charges the intermediate gateway's cut** (`6fb6cf7`, xrpl-validator) A payment book step composes its quality with `trIn` — the transfer rate of the book's IN currency when the previous step redeems — in `BookPaymentStep::adjustQualityWithFees` (BookStep.cpp:338-360), and that is what feeds `qualityUpperBound`. Offer crossing deliberately WAIVES the fee ("assume no fee is charged, or the estimate will no longer be an upper bound", BookStep.cpp:519-524); a PAYMENT does not. …
- **feat(probe): DX_FILL on the bridged legs, and lift the saturating key cap** (`8ef6ea5`, xrpl-validator) Two instruments, both inert unless their env var is set.

- **fix(xrpl-ledger): the bridged walk never crossed an expired offer either** (`2aa1274`, xrpl-validator) `live_head` — the stepper the BRIDGED walk advances with — did not test `Expiration`. The DIRECT walk has tested it since #105776250 and `reap_to_live_head` tests it too; the sibling did not. `hasExpired` is inclusive against the BASE ledger's close time (View.cpp:48; BookStep.cpp builds the stream with `sb.parentCloseTime()`), and rippled's stream collects them as removable — it logs "Removing expired offer" by …

Docs and operations

- **docs(xrpl-ledger): cite 3.2.1, not the 3.1.3 vendor tree** (`8bbbae3`, xrpl-validator)
- **docs(xrpl-ledger): renumber StrandFlow citations against 3.2.1** (`60c1d81`, xrpl-validator)

Independence program

No independence record for UTC day 2026-08-17; records begin 2026-10-20.

Network

No ledger records for this day: the reference node holds ledgers from 2026-10-04 22:52 UTC (#107,436,647).

Amendment watch

No amendment records for this day.

Open items

None.

Corrections to earlier reports

None.

XRPL research

Automated research — candidate amendment ideas

Ideas generated and prior-art-checked by the XRPL Amendment Board, an automated research tool (read-only board: iceland.halcyon-names.io). They are exploration leads, not proposals.

- **Batched NAV Redemption Windows for Vaults** (rwa) Add sfRedemptionWindowLength and sfNextRedemptionLedger SFields to the SingleAssetVault SLE, plus a bounded RedemptionQueue array of pending claims; a

- **Bounded Recurring Checks for Pull Billing** (payments) Add a tfRecurring flag plus sfFrequency (ledgers between cashes), sfNextCashableLedger, sfExpirationLedger, and sfMaxAmount SFields to CheckCreate and
- **Cross-Currency Settlement for NFT Offers** (nft) Add an optional sfSendMax SField to NFTokenCreateOffer (buy side) and route NFTokenAcceptOffer's doApply through the existing Flow payment engine inst

Appendix — commits

Committed	Merged into main	Repo	Commit	Subject
08-17 04:22	—	xrpl-validator	`04d6c8b`	fix(xrpl-ledger): a terminal issuer-only change ripples between gateways, never a book
08-17 19:57	—	xrpl-validator	`8bbbae3`	docs(xrpl-ledger): cite 3.2.1, not the 3.1.3 vendor tree
08-17 20:00	—	xrpl-validator	`60c1d81`	docs(xrpl-ledger): renumber StrandFlow citations against 3.2.1
08-17 22:21	—	xrpl-validator	`23f1153`	fix(xrpl-ledger): the balance clamps the GROSS, and a pass that misses limitQuality is discarded whole
08-18 01:26	—	xrpl-validator	`8e48764`	fix(xrpl-ledger): a pseudo-account cannot receive an escrow
08-18 02:29	—	xrpl-validator	`6fb6cf7`	fix(xrpl-ledger): the strand upper bound charges the intermediate gateway's cut
08-18 02:59	—	xrpl-validator	`8ef6ea5`	feat(probe): DX_FILL on the bridged legs, and lift the saturating key cap
08-18 03:24	—	xrpl-validator	`2aa1274`	fix(xrpl-ledger): the bridged walk never crossed an expired offer either

Times in UTC. A commit appears on the day it was committed to main or merged into it.